THE UNIVERSITY OF TEXAS AT DALLAS

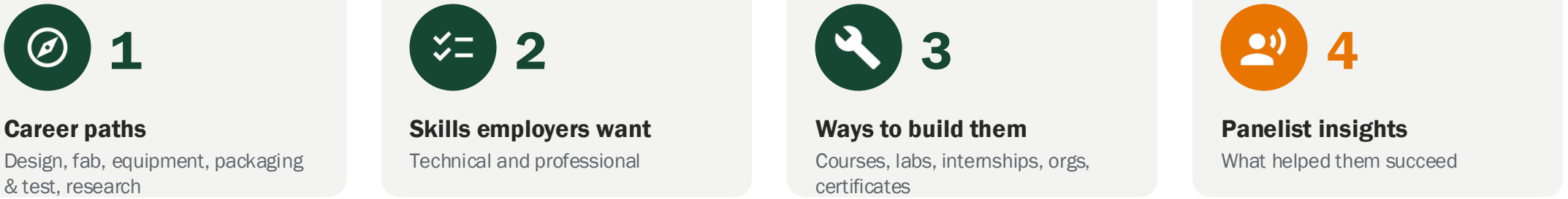
ECS 1100

Sector: Semiconductor & Microelectronics Jobs

Learning Objectives

By the end of the week, you will be able to:

1. Identify career paths in the semiconductor and microelectronics industry across sectors such as design, fabrication, equipment, packaging and testing, and research.
2. Describe the technical and professional skills employers in these sectors look for.
3. Outline ways to build those skills through coursework, research labs, internships, student organizations, and certifications.
4. Summarize key insights from the industry panelists about their career journeys and what helped them succeed.



What is a semiconductor?

A semiconductor is a material, most often silicon, whose ability to carry current sits between a metal and an insulator. Engineers tune that ability by adding traces of other elements (doping) and by applying a voltage.

A transistor uses that control as a switch with no moving parts. Put billions of them on one sliver of silicon and you have an integrated circuit, or chip.

Microelectronics is the field that designs, fabricates, tests and packages those chips, and builds the equipment, software and data systems that make them.

Where a material sits on the conductivity scale

Insulator	Semiconductor	Conductor
glass, plastic, oxide	silicon, SiC, GaN, GaAs	copper, aluminum

Where chips end up

Phones & PCs	Cars & EVs	AI data centers
Medical devices	Power & grid	Factory robots



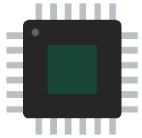
Sand
Mostly silicon dioxide



Ingot
Purified, grown as one crystal



Wafer
Sliced thin; holds hundreds of chips



Chip
One die, cut out and packaged

How a chip gets made

Design

Architecture, circuits, layout, verification

Majors often here

EE ME CS

Wafer fab

Lithography, deposition, etch, doping, CMP, metrology

Majors often here

MSE ME EE

Test

Probe and final test, yield and failure analysis

Majors often here

EE ME DS

Packaging

Dicing, bonding, substrates, thermal design

Majors often here

ME MSE EE

Application

Boards, firmware, customer support

Majors often here

EE CE BME











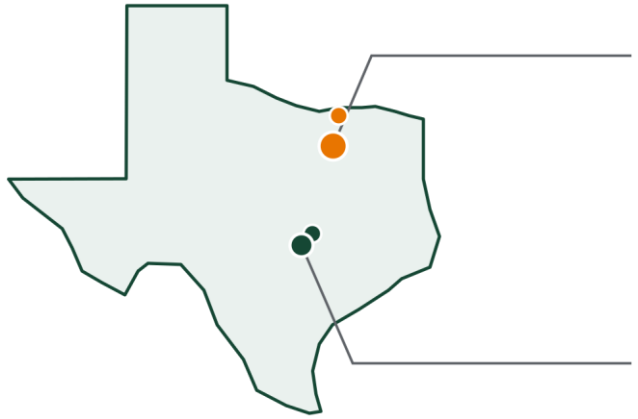
Why North Texas is a semiconductor hub

74
of 180 postings are in the DFW area

50
more are elsewhere in Texas

69%
of all postings are in Texas

- **Texas Instruments** is headquartered in Dallas and runs 300 mm fabs in Richardson and Sherman. It is the largest employer in this job log (46 postings).
- **Qorvo** has major RF chip operations in Richardson.
- **Central Texas** adds Samsung (Austin and Taylor), NXP and AMD.
- **UT Dallas** hosts the North Texas Semiconductor Institute (NTxSI), launched in 2023, and leads the South regional node of the National Network for Microelectronics Education.



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major expectations

What every employer expects, whatever your major

1

Teamwork

Work across design, fab, test and customer teams. Nearly every posting says it.

2

Communication

Write clear reports and specs, and explain results to people outside your field.

3

Problem solving

Find the root cause of a failure, then prove the fix with data.

4

Data and scripting

Basic statistics and a scripting language, usually Python, to handle test and fab data.

These four come up in almost every posting, for every major. Teamwork alone appears in 96% of the 180 postings in the Skills Matrix section.

Major expectations

Circuits, chips and devices

EE Electrical Engineering

Companies ask for

- Analog, digital and RF circuit fundamentals
- Device physics: how transistors behave
- Lab work with scopes, meters and analyzers
- Scripting to automate tests and sort data

Typical entry roles

Design, verification, product, test, device and reliability engineer

CE Computer Engineering

Companies ask for

- Digital logic and computer architecture
- Verilog / SystemVerilog, RTL and verification
- C/C++, Python and Linux
- Test development and validation

Typical entry roles

Digital design, design verification, DFT, embedded and validation engineer

BME Biomedical Engineering

Companies ask for

- Analog and mixed-signal basics
- Bench work with LabVIEW and MATLAB
- Schematics and PCB reading
- Sensing and medical chip applications

Typical entry roles

Applications engineer, biomedical or sensing intern (fewer direct roles)

Major expectations

Fab, materials and equipment

MSE Materials Science & Engineering

Companies ask for

- Fab processes: litho, thin films, etch, CMP
- Materials and electrical characterization
- Statistics, DOE, SPC and yield
- Cleanroom work and root cause analysis

Typical entry roles

Process, integration, metrology, yield and packaging materials engineer

ME Mechanical Engineering

Companies ask for

- Equipment troubleshooting and upkeep
- CAD, tolerancing, thermal and FEA
- Vacuum, pneumatics and controls
- Packaging mechanics and lean methods

Typical entry roles

Equipment, field service, packaging, manufacturing and facilities engineer

SYSE Systems Engineering

Companies ask for

- Statistics and data analysis
- Capacity modeling, scheduling, operations research
- Lean and continuous improvement
- Quality systems

Typical entry roles

Industrial engineer, operations research analyst, quality engineer

Major expectations

Software, data and AI

CS Computer Science

Companies ask for

- Python, scripting and automation
- Machine learning and computer vision for inspection
- Data engineering, SQL and cloud
- C/C++ and Linux for EDA and embedded work

Typical entry roles

Fab automation software engineer, algorithm or AI engineer, EDA infrastructure

SE Software Engineering

Companies ask for

- Software for fabs and chip design
- DevOps and CI for EDA tools
- Firmware
- Data tools and SQL

Typical entry roles

Software or firmware engineer, DevOps engineer, data engineer

DS Data Science

Companies ask for

- Statistics, Python and SQL
- Machine learning and anomaly detection
- Dashboards: Tableau, Power BI, JMP, Spotfire
- Yield and defect analytics

Typical entry roles

Data analyst, data engineer, data scientist for yield and defects

All Jobs are Posted

go to this website to answer to

Your ICP and HW

CLICK HERE



ICP A link: ECS 1100 Semiconductor/Microelectronics Sector Panel - ICP.1 – Fill out form

How the matrix was built

01 **Collect**

180 postings open on Sept 26, 2026:

02 **Match**

About 60 skill terms searched in each posting's text

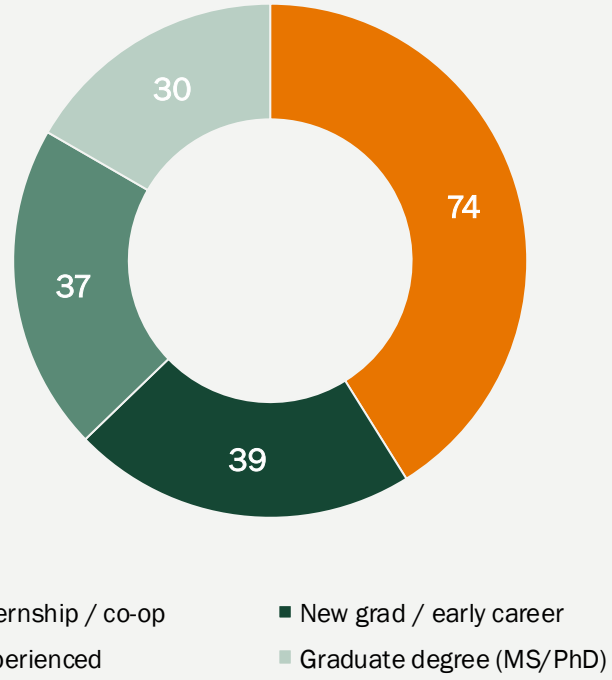
03 **Count**

A skill counts once per posting

04 **Split**

Major columns show the share of that major's postings that mention the skill. The idea of one posting can fit several majors was taken into account.

Career stage of the 180 postings



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Skills Matrix

Nearly every posting asks for teamwork

96%
Teamwork & cross-functional collaboration

82%
Communication (written & verbal)

62%
Problem solving & analytical thinking

47%
Documentation & specifications

Teamwork, by major

EE	CE	MSE	ME	CS	SE	DS	SYSE	BME
97%	97%	97%	98%	93%	93%	89%	94%	100%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

The 15 most requested technical skills



47%

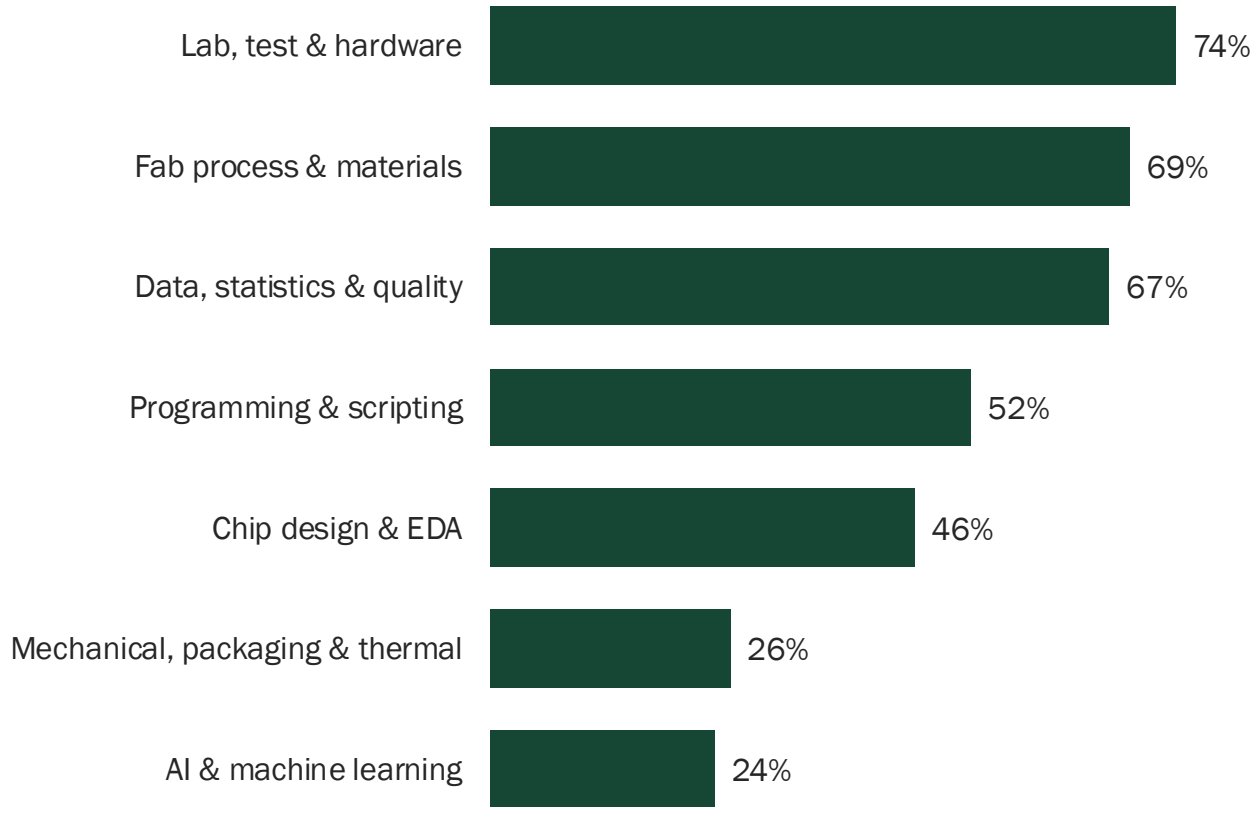
ask for statistics & data analysis, the most requested technical skill

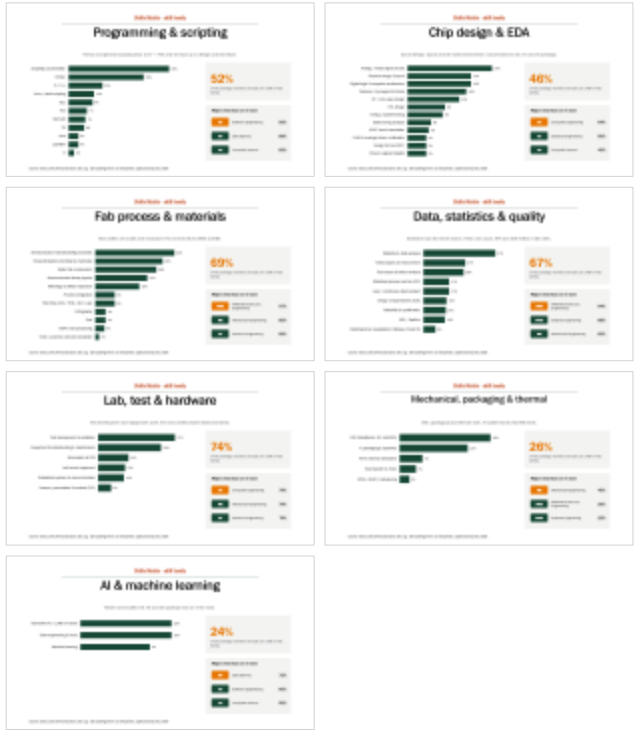
39% · 29%

name scripting & automation and Python, across hardware and fab roles alike

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Seven technical skill families

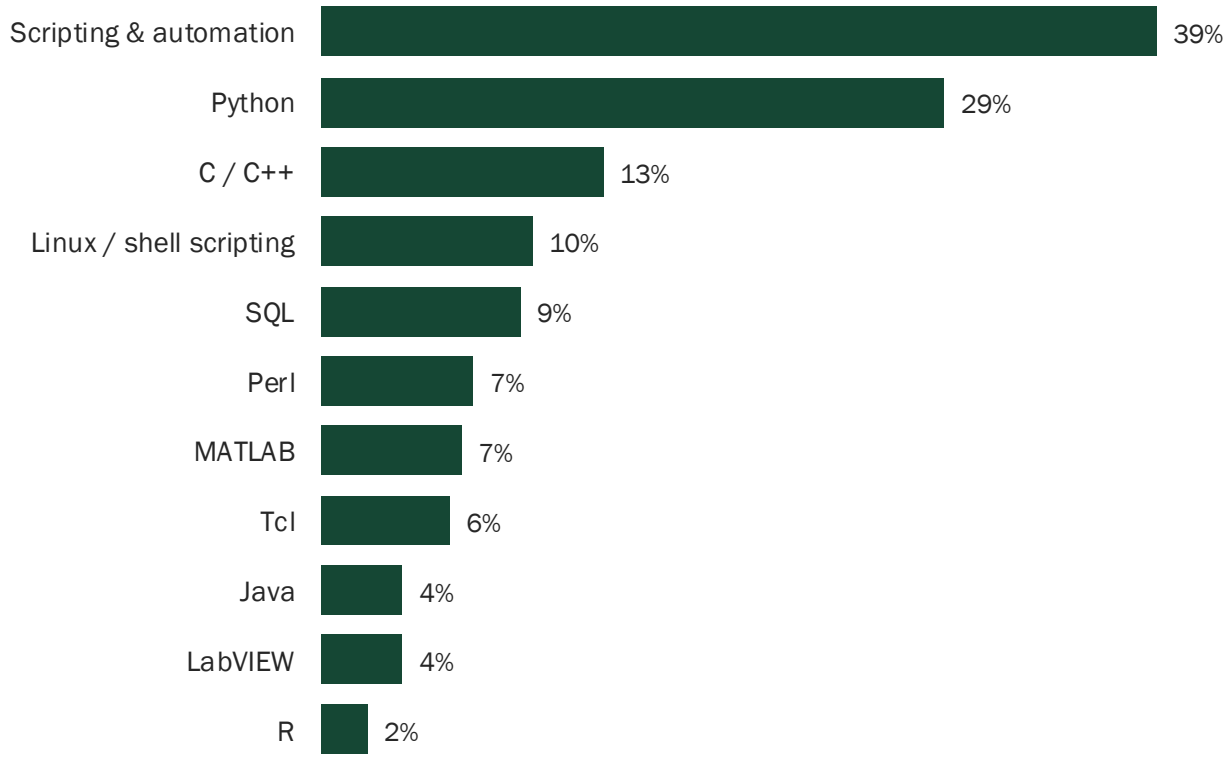




Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Programming & scripting

Python and general scripting lead. C/C++, Perl and Tcl show up in design and test flows.



52%

of all postings mention at least one skill in this family

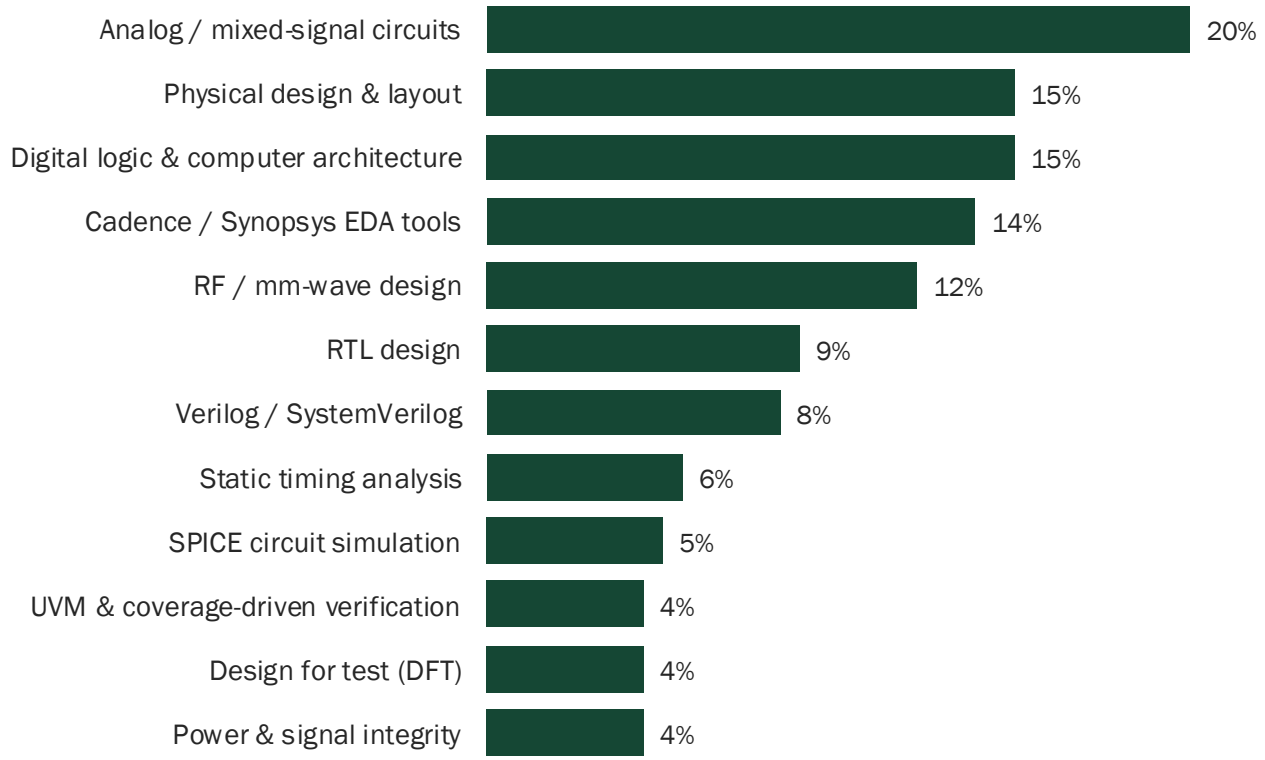
Majors that lean on it most

SE	Software Engineering	93%
DS	Data Science	89%
CS	Computer Science	83%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Chip design & EDA

Circuit design, layout and the tools behind them. Concentrated in CE, EE and CS postings.



46%

of all postings mention at least one skill in this family

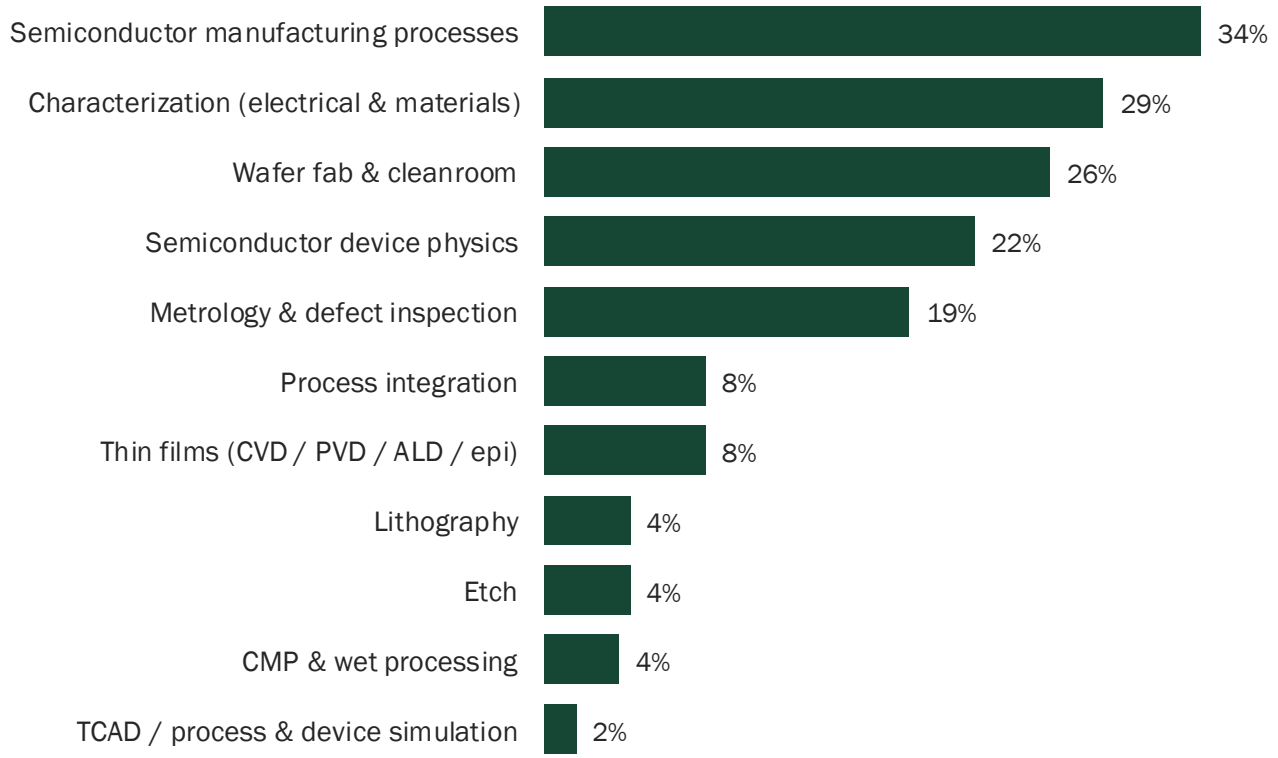
Majors that lean on it most

CE	Computer Engineering	64%
EE	Electrical Engineering	50%
CS	Computer Science	43%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Fab process & materials

How wafers are made and measured. The core family for MSE and ME.



69%

of all postings mention at least one skill in this family

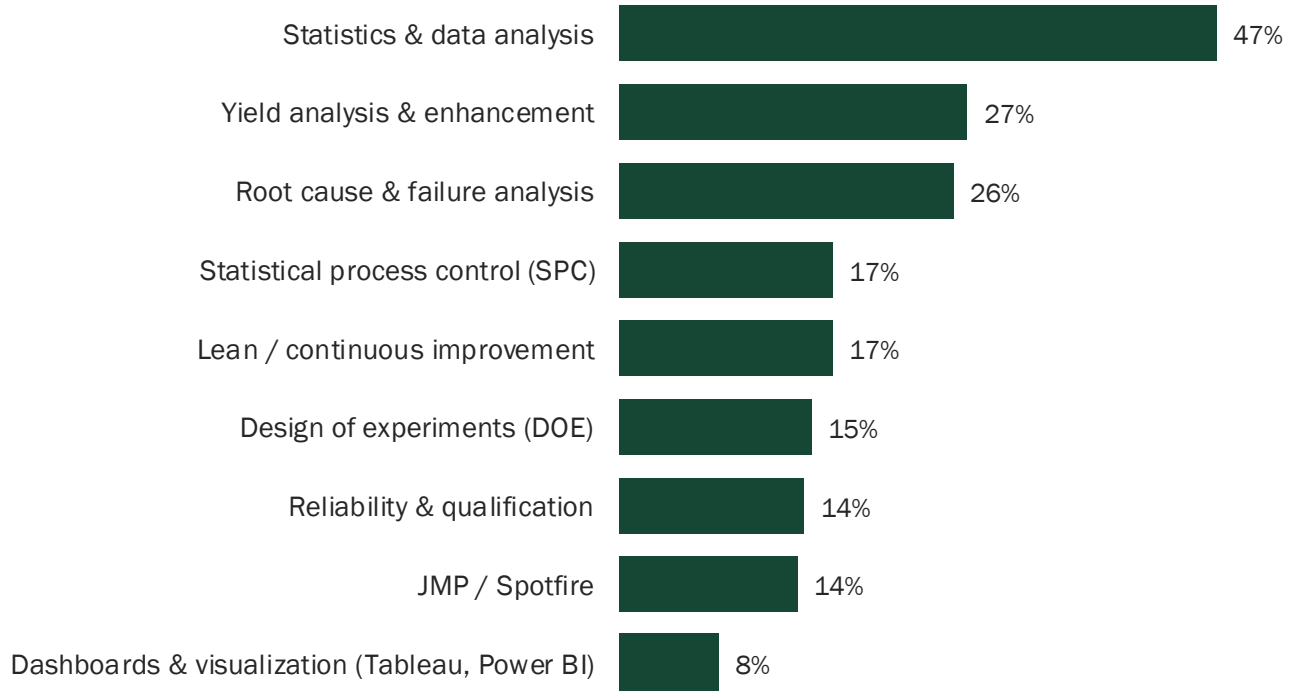
Majors that lean on it most

MSE	Materials Science & Engineering	97%
ME	Mechanical Engineering	85%
EE	Electrical Engineering	69%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Data, statistics & quality

Statistics tops the whole matrix. Yield, root cause, SPC and DOE follow in fab roles.



67%

of all postings mention at least one skill in this family

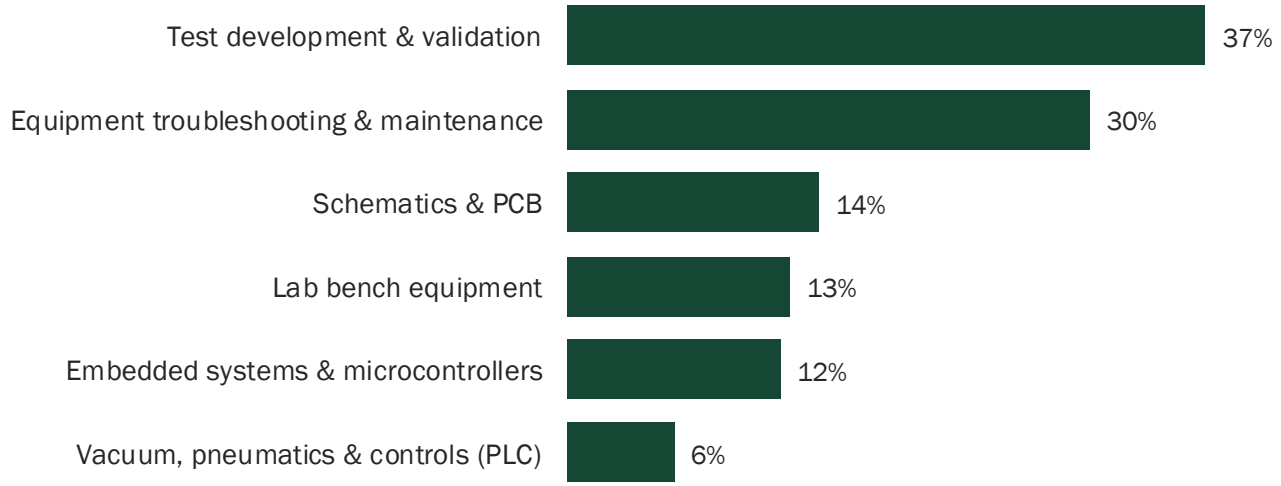
Majors that lean on it most

MSE	Materials Science & Engineering	94%
SYSE	Systems Engineering	88%
ME	Mechanical Engineering	83%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Lab, test & hardware

Test development and equipment work, the most widely shared technical family.



74%

of all postings mention at least one skill in this family

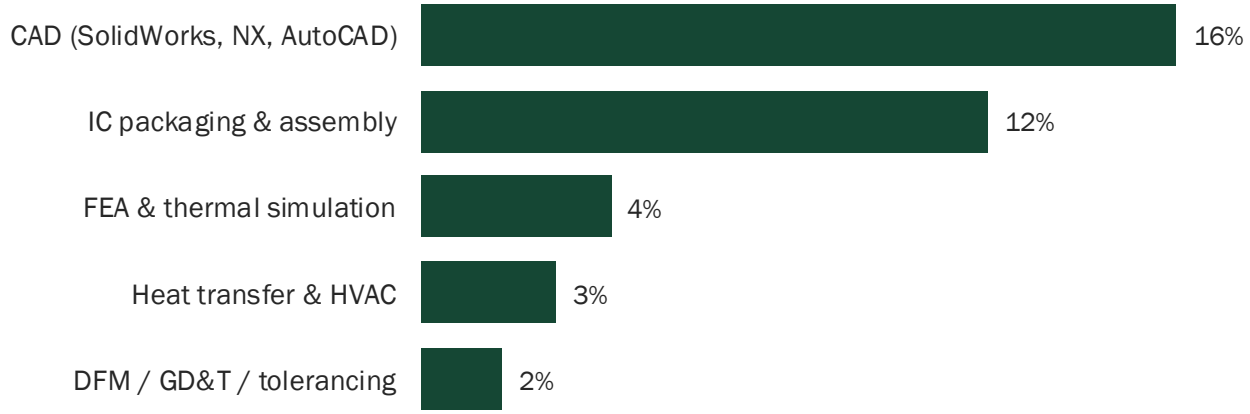
Majors that lean on it most

CE	Computer Engineering	79%
ME	Mechanical Engineering	78%
EE	Electrical Engineering	74%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Mechanical, packaging & thermal

CAD, packaging and thermal work. A smaller family that ME leads.



26%

of all postings mention at least one skill in this family

Majors that lean on it most

ME	Mechanical Engineering	45%
MSE	Materials Science & Engineering	26%
SYSE	Systems Engineering	25%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

AI & machine learning

Newer and smaller. DS, SE and CS postings lean on it the most.



24%

of all postings mention at least one skill in this family

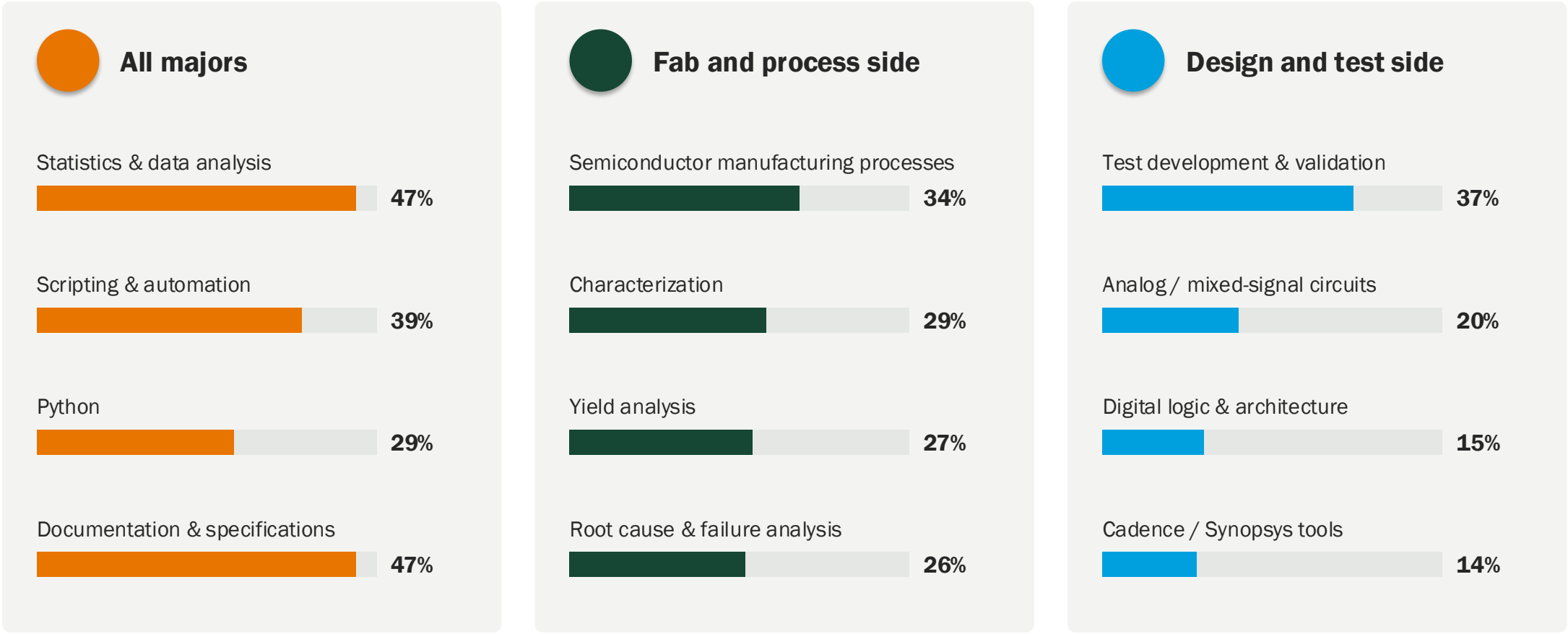
Majors that lean on it most

DS	Data Science	72%
SE	Software Engineering	60%
CS	Computer Science	50%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Skills Matrix

What the matrix suggests building first



Bars show share of all 180 postings (full bar = 50%)

Skills × Majors

The whole picture in one grid

Share of each major's postings that mention at least one skill from each family

Skill family	All	EE	CE	MSE	ME	CS	SE	DS	SYSE	BME*
postings →	180	151	75	62	60	46	15	18	16	3
Professional skills	99%	99%	99%	100%	100%	98%	93%	94%	100%	100%
Lab, test & hardware	74%	74%	79%	68%	78%	67%	67%	39%	56%	100%
Fab process & materials	69%	69%	52%	97%	85%	57%	60%	61%	69%	·
Data, statistics & quality	67%	69%	49%	94%	83%	61%	40%	83%	88%	33%
Programming & scripting	52%	53%	75%	34%	35%	83%	93%	89%	62%	100%
Chip design & EDA	46%	50%	64%	18%	20%	43%	40%	17%	31%	67%
Mechanical, packaging & thermal	26%	25%	17%	26%	45%	7%	13%	6%	25%	·
AI & machine learning	24%	24%	36%	15%	12%	50%	60%	72%	19%	33%

Share of the major's postings: 0% | 25% | 50% | 75% | 100%+

*BME has only 3 postings

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Skills × Majors

Top 20 technical skills by major

Skill	All	EE	CE	MSE	ME	CS	SE	DS	SYSE	BME*
postings →	180	151	75	62	60	46	15	18	16	3
Statistics & data analysis	47%	50%	29%	76%	57%	52%	33%	72%	69%	33%
Scripting & automation	39%	39%	53%	26%	30%	61%	67%	72%	56%	·
Test development & validation	37%	38%	55%	21%	27%	37%	20%	28%	38%	·
Semiconductor manufacturing processes	34%	32%	19%	63%	47%	35%	27%	33%	50%	·
Equipment troubleshooting & maintenance	30%	26%	15%	42%	50%	24%	27%	11%	19%	·
Python	29%	30%	43%	21%	15%	59%	67%	72%	25%	33%
Characterization (electrical & materials)	29%	30%	21%	45%	30%	15%	·	·	6%	·
Yield analysis & enhancement	27%	29%	16%	47%	32%	26%	20%	17%	31%	·
Wafer fab & cleanroom	26%	25%	8%	50%	37%	15%	13%	11%	31%	·
Root cause & failure analysis	26%	25%	15%	44%	42%	24%	13%	22%	25%	·
Semiconductor device physics	22%	26%	15%	32%	20%	13%	7%	11%	·	·
Analog / mixed-signal circuits	20%	23%	27%	2%	3%	11%	·	6%	·	67%
Metrology & defect inspection	19%	19%	8%	40%	30%	11%	27%	17%	12%	·
Statistical process control (SPC)	17%	19%	5%	34%	22%	13%	13%	11%	6%	·
Lean / continuous improvement	17%	15%	11%	19%	28%	11%	7%	17%	38%	33%
CAD (SolidWorks, NX, AutoCAD)	16%	15%	13%	5%	18%	4%	13%	·	12%	·
Physical design & layout	15%	17%	21%	3%	2%	11%	7%	6%	12%	·
Digital logic & computer architecture	15%	17%	33%	·	·	26%	7%	6%	·	·
Design of experiments (DOE)	15%	16%	1%	37%	28%	·	·	·	·	·
Reliability & qualification	14%	15%	7%	19%	20%	7%	·	·	12%	·

**BME has only 3 postings*

Share of the major's postings: 0% | 20% | 40% | 60% | 80%+

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Skills ranked 21 to 40 by major

Skill	All	EE	CE	MSE	ME	CS	SE	DS	SYSE	BME*
postings →	180	151	75	62	60	46	15	18	16	3
Schematics & PCB	14%	16%	17%	5%	12%	7%	13%	·	·	67%
Cadence / Synopsys EDA tools	14%	15%	24%	·	·	13%	20%	6%	·	·
JMP / Spotfire	14%	15%	3%	29%	20%	15%	7%	11%	19%	·
C / C++	13%	16%	29%	·	2%	28%	20%	6%	6%	67%
Lab bench equipment	13%	15%	20%	·	5%	13%	7%	·	·	67%
RF / mm-wave design	12%	14%	3%	10%	15%	7%	13%	6%	12%	·
Embedded systems & microcontrollers	12%	12%	17%	8%	10%	17%	27%	6%	·	33%
IC packaging & assembly	12%	13%	7%	23%	25%	2%	·	6%	6%	·
Generative AI / LLMs & AI tools	12%	11%	13%	13%	7%	24%	27%	33%	19%	·
Data engineering & cloud	12%	11%	21%	2%	2%	24%	33%	39%	6%	·
Linux / shell scripting	10%	10%	21%	·	·	26%	27%	·	·	·
SQL	9%	9%	9%	11%	7%	24%	40%	50%	12%	·
RTL design	9%	11%	20%	·	·	9%	·	·	·	·
Machine learning	9%	8%	12%	5%	3%	30%	40%	56%	6%	33%
Verilog / SystemVerilog	8%	10%	19%	·	·	13%	·	·	·	·
Process integration	8%	9%	5%	15%	5%	7%	13%	6%	·	·
Thin films (CVD / PVD / ALD / epi)	8%	8%	3%	18%	12%	4%	7%	6%	·	·
Dashboards & visualization (Tableau, Power BI)	8%	7%	8%	10%	8%	13%	13%	39%	19%	·
Perl	7%	9%	17%	2%	2%	7%	·	·	·	·
MATLAB	7%	7%	7%	10%	10%	11%	·	11%	19%	33%

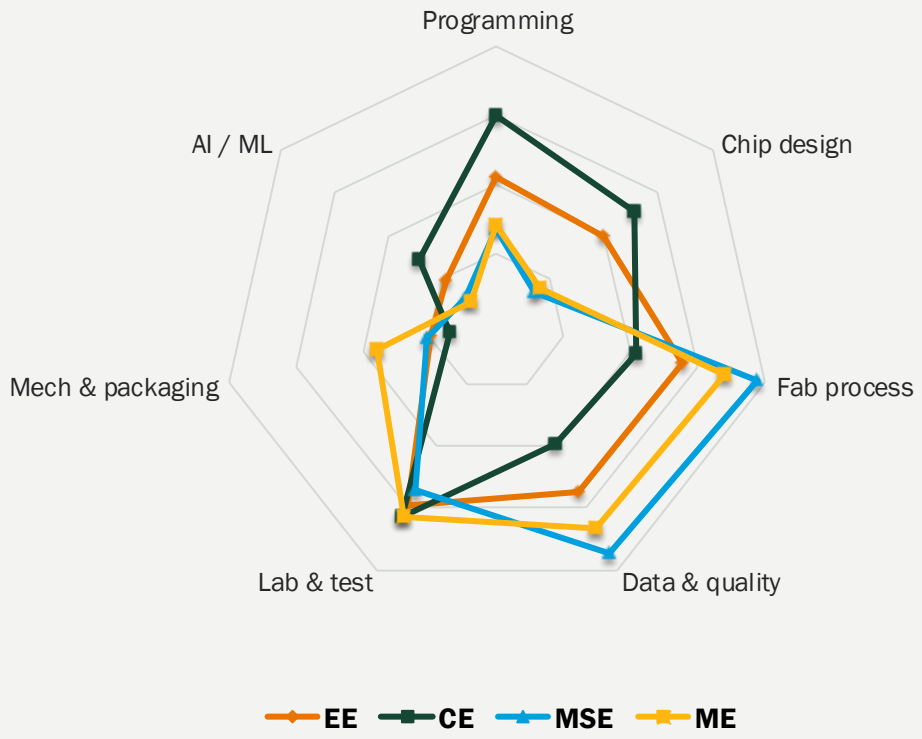
**BME has only 3 postings*

Share of the major's postings: 0% | 20% | 40% | 60% | 80%+

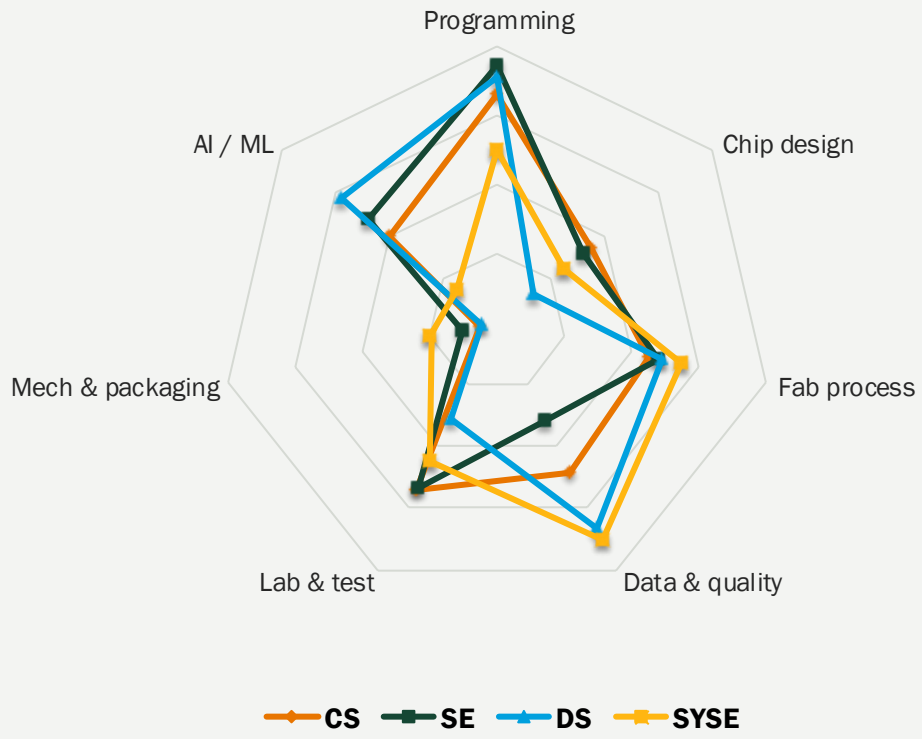
Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Skill priority distribution

Hardware & materials majors



Computing & systems majors



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Career support on campus

Jonsson School Career Services

One-on-one help with resumes, cover letters, mock interviews and LinkedIn profiles, plus internship approvals, including CPT.

jonssonschooljobs@utdallas.edu

Internship Fair and 12twenty

The Jonsson School runs its own internship fair; 12twenty has a UTD portal.

https://utdallas.12twenty.com/ .

UTDesign capstone

Industry-sponsored senior design projects. A chance to work on a real company problem before graduation.

utdesign.utdallas.edu

LinkedIn Learning (free)

12,000+ video courses through McDermott Library with your NetID: Python, MATLAB, Excel, SolidWorks, statistics. Access ends at graduation.

library.utdallas.edu › LinkedIn Learning

Sources: engineering.utdallas.edu, career.utdallas.edu, library.utdallas.edu (Sept 2026)

Free certificates: Coursera + UT Dallas

What you get

- Free access to Coursera Career Academy for every UTD student
- 35+ professional certificates from Google, IBM, Microsoft and others
- Certificates cost nothing and can be shared on LinkedIn
- Sign up with your @utdallas.edu email from the Provost's Coursera page
- Coursera Plus content outside Career Academy is not included

Questions: microcredentials@utdallas.edu

Match certificates to the skills employers ask for

Skill	Postings	Certificate to look for
Statistics & data analysis	**47%**	Google Data Analytics · IBM Data Analyst
Scripting & Python	**39%**	Google IT Automation with Python, Automation and Scripting with Python (Microsoft)
Lean / continuous improvement	**17%**	Google Project Management, Generative AI for Project Managers Specialization (IBM)
AI & machine learning	**24%***	Google Advanced Data Analytics · IBM AI Engineering
Dashboards & visualization	**8%**	Microsoft Power BI Data Analyst · Google Business Intelligence

Sources: provost.utdallas.edu, UTD Coursera FAQ, utsystem.edu (Sept 2026)

Semiconductor-specific resources

North Texas Semiconductor Institute

Events, an industry FAQ and job listings. Its Semiconductor Scholar Program funds MS students in MSE, EE or ME (U.S. citizens or permanent residents, 3.0+ GPA).

ntxsi.utdallas.edu

Cleanroom Research Laboratory

Hands-on fab tools. Access goes through a faculty sponsor, online safety modules, a walkthrough and tool training, so joining a research lab is the way in.

cores.research.utdallas.edu/cleanroom

Nanoscience and Technology minor

Offered by Materials Science & Engineering. Adds nanomaterials and device fundamentals to any engineering major.

UTD undergraduate catalog › ECS minors

Student organizations

IEEE UTD, the IEEE Solid-State Circuits Society student chapter, SAMPE, ASME, SWE, NSBE, SHPE, SASE and the AI Society.

engineering.utdallas.edu › student organizations

Sources: ntxsi.utdallas.edu, UTD cleanroom site, UTD catalog (Sept 2026)

APPENDIX

MAJOR GUIDE

Major Guide

Electrical Engineering (EE)

151 postings fit

61 internships / co-ops

36 new grad / early

65 in the DFW area

What they do

Analog/digital IC design, design verification, RF design, product & test engineering, device and process engineering, reliability.

Entry titles seen

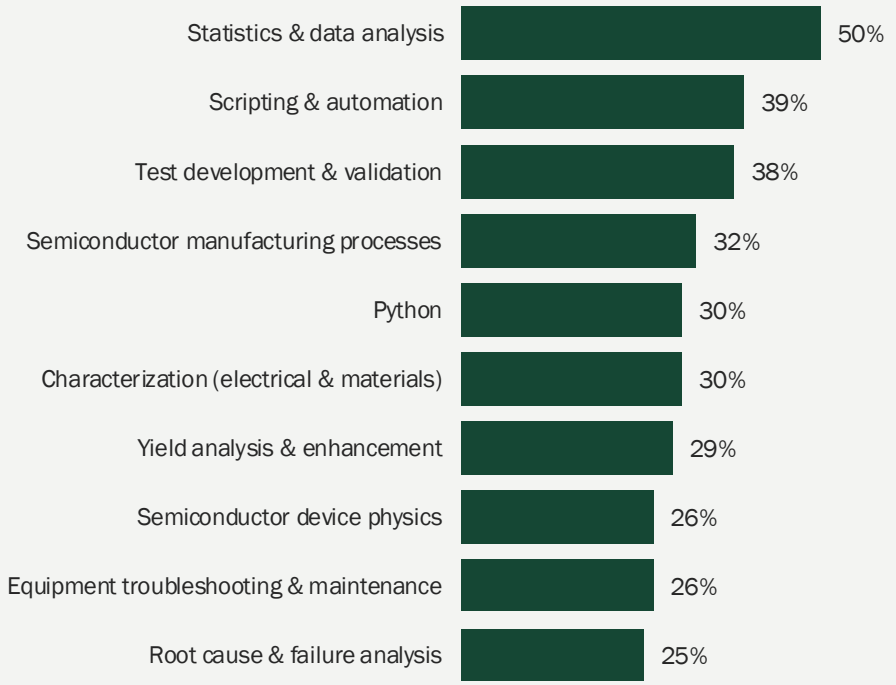
- Analog IC Design Engineering Intern
- Associate Design Verification Engineer
- NCG Mixed Signal Design Engineer (HBM)
- Product Engineering Intern

Where it leads in 5 to 8 years

- HBM Power Delivery Network Engineer
- Wide Bandgap Device/PI Engineer
- Senior Physical Design Engineer

Most repeated technical skills

Share of the 151 EE postings that mention each skill



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major Guide

Computer Engineering (CE)

75 postings fit

37 internships / co-ops

17 new grad / early

31 in the DFW area

What they do

Digital design and verification, DFT, embedded firmware, product development test programs, data engineering.

Entry titles seen

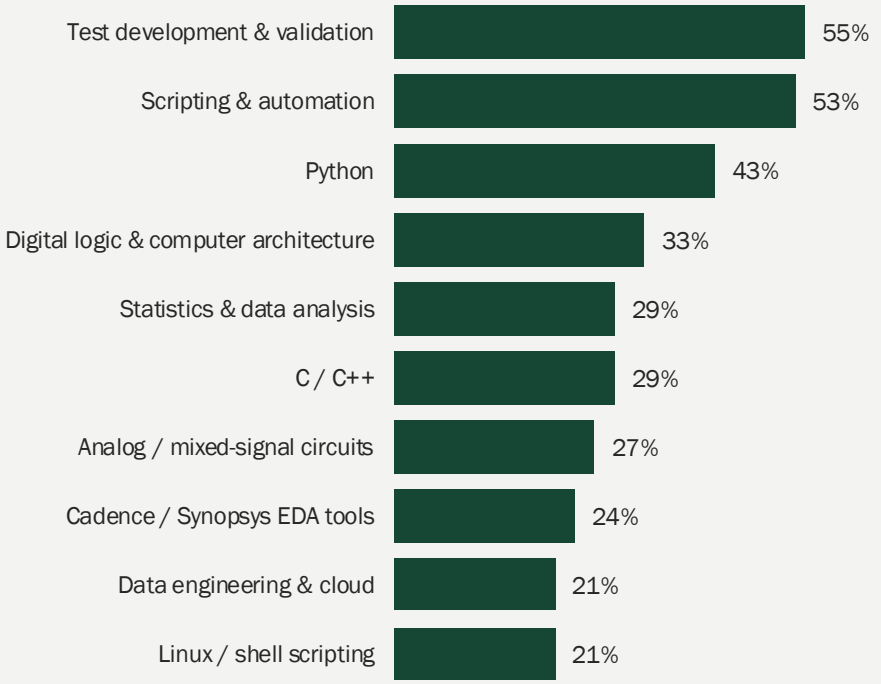
- Digital IC Design Engineering Intern
- Hardware Design Verification Intern (AMD)
- ASIC Verification Engineer - NCG

Where it leads in 5 to 8 years

- Senior Physical Design Engineer
- Yield & Process Control Automation Engineer

Most repeated technical skills

Share of the 75 CE postings that mention each skill



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major Guide

Materials Science & Engineering (MSE)

62 postings fit

22 internships / co-ops

12 new grad / early

19 in the DFW area

What they do

Process development and integration, lithography, thin films, packaging materials, metrology, defect and yield.

Entry titles seen

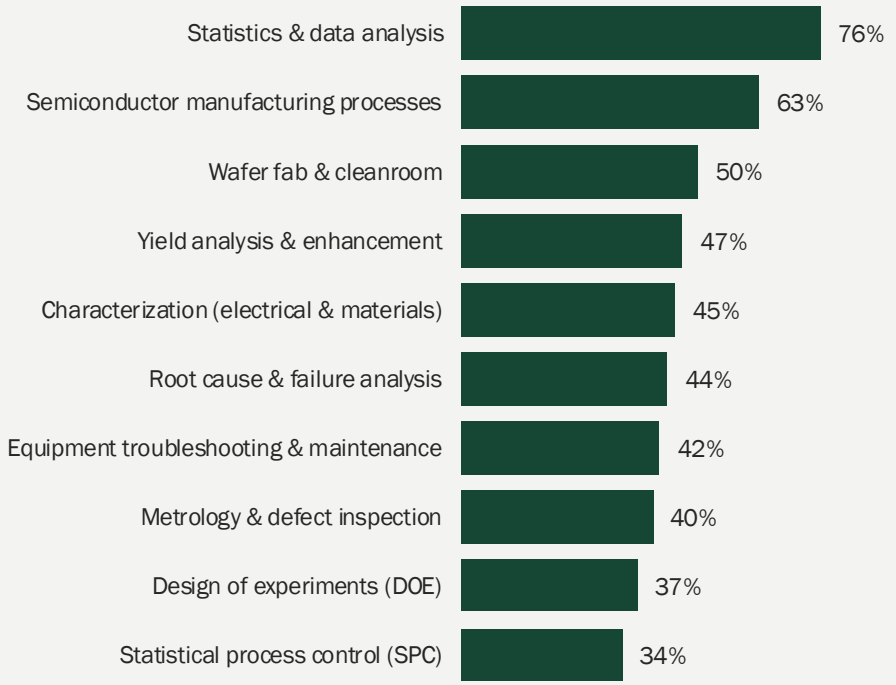
- Process Development Engineer Intern (MS/PhD)
- Intern - Photolithography Process
- NCG Photolithography Process Engineer

Where it leads in 5 to 8 years

- Process Development Engineer - Advanced Lithography
- Wide Bandgap Device/PI Engineer
- Device Engineer (yield)

Most repeated technical skills

Share of the 62 MSE postings that mention each skill



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major Guide

Mechanical Engineering (ME)

60 postings fit

25 internships / co-ops

14 new grad / early

23 in the DFW area

What they do

Equipment engineering, field service, packaging mechanics and thermal, fab facilities (HVAC, cooling), manufacturing engineering.

Entry titles seen

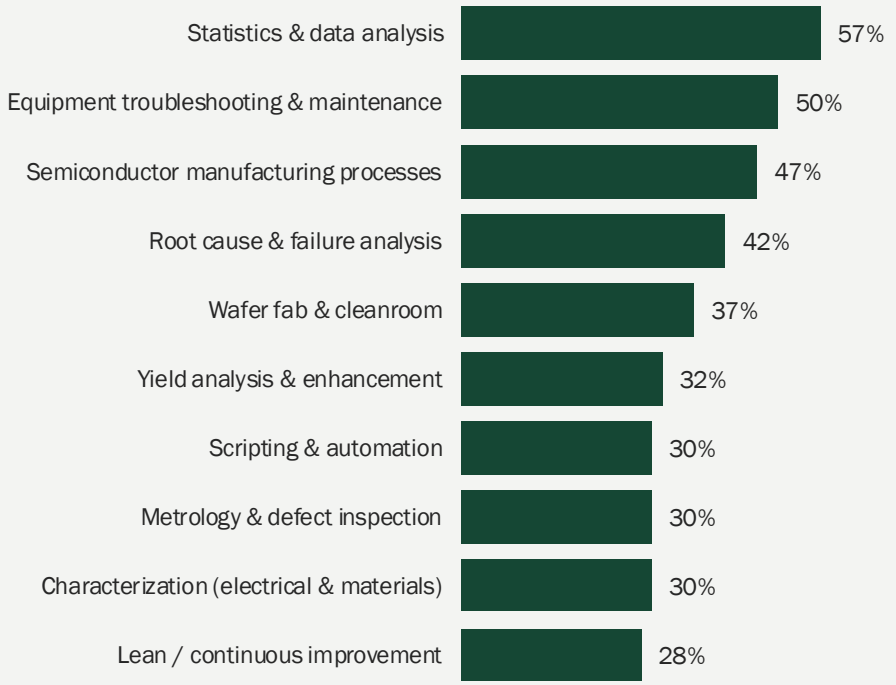
- Mechanical Engineering Intern (Qorvo)
- Summer 2027 ME Intern (AMAT)
- Field Service Engineer NCG Rotational Program

Where it leads in 5 to 8 years

- Engineer, Semi Packaging
- CMP Manufacturing Equipment Engineer
- HVAC Facilities Thermal Engineer

Most repeated technical skills

Share of the 60 ME postings that mention each skill



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major Guide

Computer Science (CS)

46 postings fit

20 internships / co-ops

12 new grad / early

14 in the DFW area

What they do

Fab data and automation software, AI/computer vision for inspection, embedded software, EDA infrastructure.

Entry titles seen

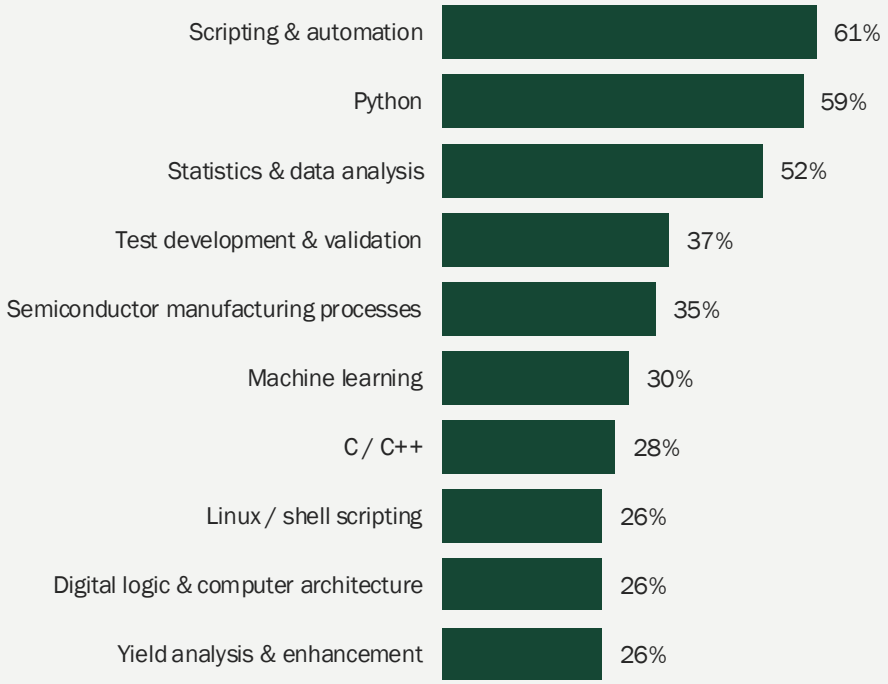
- Smart Manufacturing and Automation Intern
- CAP Software Engineer
- Algorithm Engineering Intern (AI/CV)

Where it leads in 5 to 8 years

- Sr AI Architect - Yield & Process Control
- Senior Data Scientist, Defect Quality

Most repeated technical skills

Share of the 46 CS postings that mention each skill



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major Guide

Software Engineering (SE)

15 postings fit

3 internships / co-ops

3 new grad / early

9 in the DFW area

What they do

Software for fabs and chip design: automation, DevOps for EDA tools, firmware, data tools.

Entry titles seen

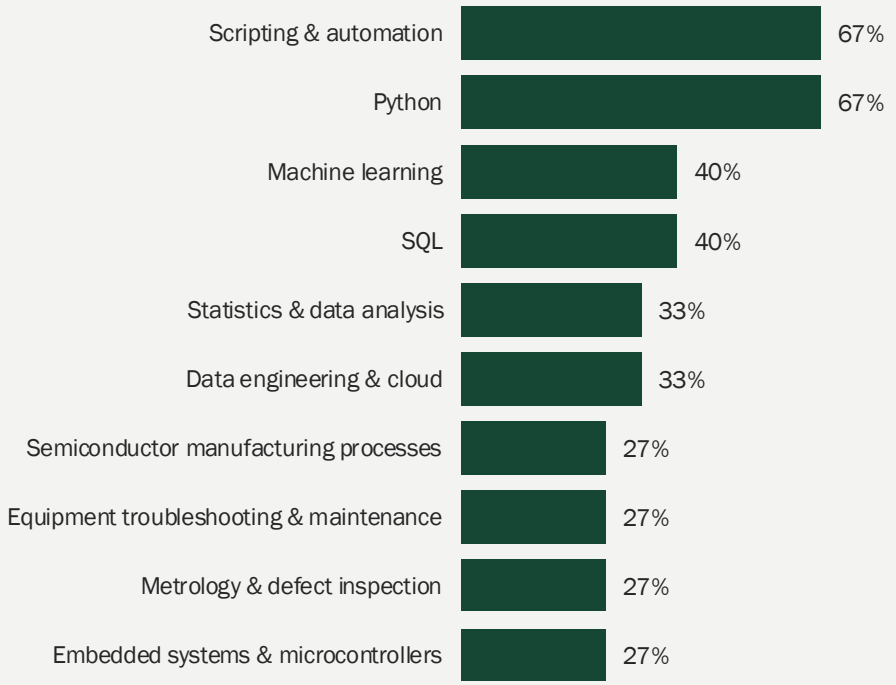
- Software-Firmware Engineering Intern
- CAP Software Engineer
- CAP Data Engineer

Where it leads in 5 to 8 years

- DevOps Engineer - Semiconductor EDA Tools
- Yield & Process Control Automation Engineer

Most repeated technical skills

Share of the 15 SE postings that mention each skill



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major Guide

Data Science (DS)

18 postings fit

11 internships / co-ops

2 new grad / early

5 in the DFW area

What they do

Yield and defect analytics, dashboards, manufacturing data engineering, ML for anomaly detection.

Entry titles seen

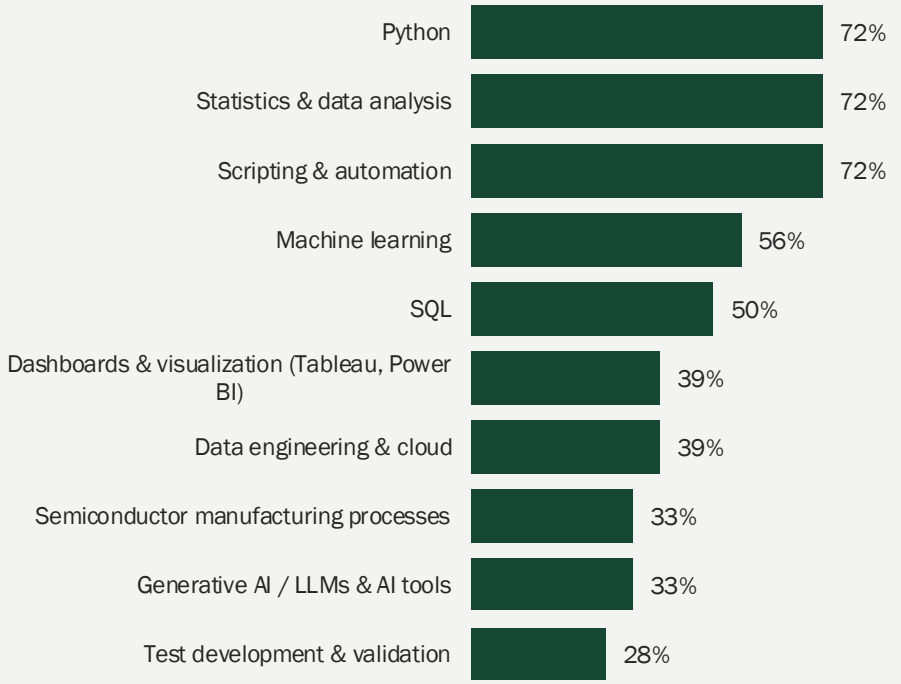
- Data Analyst Intern (AMD)
- CAP Data Engineer
- Smart Manufacturing and Automation Intern

Where it leads in 5 to 8 years

- Senior Data Scientist, Defect Quality & Systems
- Sr AI Architect - Yield & Process Control

Most repeated technical skills

Share of the 18 DS postings that mention each skill



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major Guide

Systems Engineering (SYSE)

16 postings fit

9 internships / co-ops

6 new grad / early

5 in the DFW area

What they do

Fab capacity modeling, scheduling and operations research, quality systems, platform/system engineering.

Entry titles seen

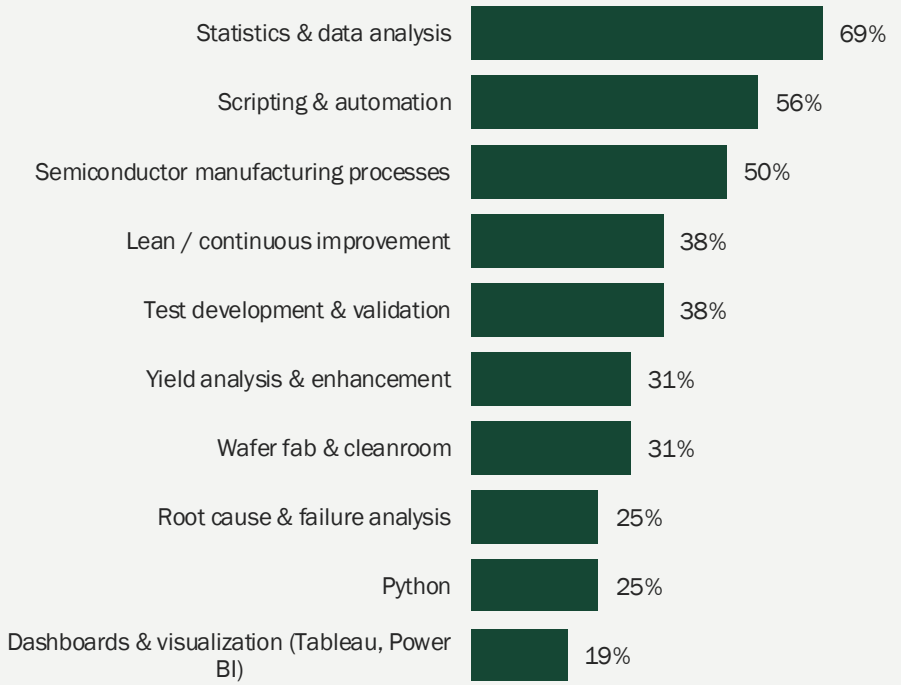
- North Texas Industrial Engineer Intern (TI, names Systems Engineering)
- CAP Industrial Engineer (TI)
- Operations Research Graduate Intern (Intel)
- Analytical Quality Engineering Co-Op (Entegris)

Where it leads in 5 to 8 years

- Factory-planning, operations and quality leadership (few senior examples in this set)

Most repeated technical skills

Share of the 16 SYSE postings that mention each skill



Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

Major Guide

Biomedical Engineering (BME)

3 postings fit

2 internships / co-ops

1 new grad / early

2 in the DFW area

What they do

Few direct semiconductor roles. The doors in are applications engineering (TI names Bio-Medical Engineering) and biosensing chips at ADI.

Entry titles seen

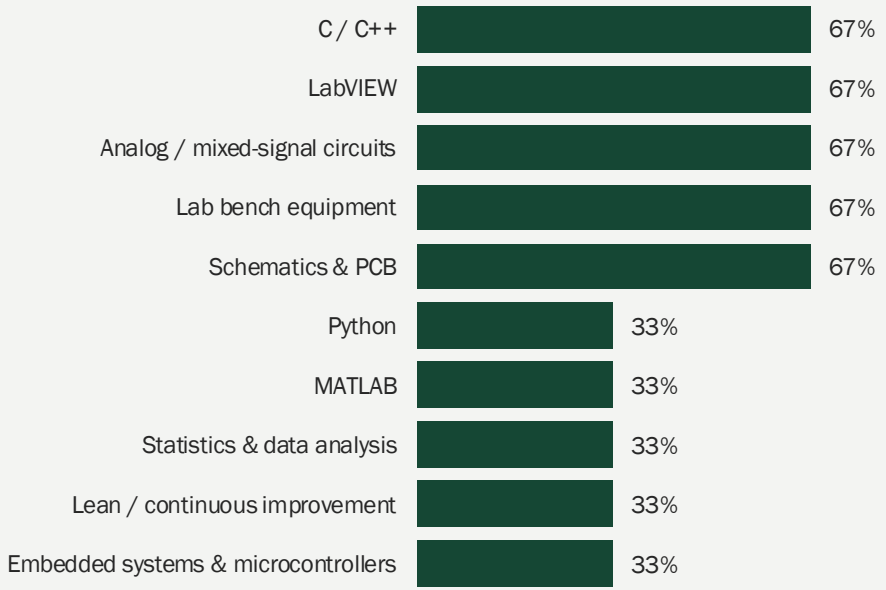
- Applications Engineering Intern (TI)
- CAP Applications Engineer (TI)
- Biomedical Engineering Intern (Analog Devices)

Where it leads in 5 to 8 years

- Applications and systems roles for medical and sensing chips

Most repeated technical skills

Share of the 3 BME postings that mention each skill



Only 3 postings fit BME, so read these as examples rather than trends.

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

APPENDIX: Skills × Majors

Signature skill for each major

EE — Semiconductor device physics

EE postings	26%
all others	0%

CE — Digital logic & computer architecture

CE postings	33%
all others	2%

MSE — Statistics & data analysis

MSE postings	76%
all others	31%

ME — Equipment troubleshooting & maintenance

ME postings	50%
all others	20%

CS — Python

CS postings	59%
all others	19%

SE — Python

SE postings	67%
all others	26%

DS — Machine learning

DS postings	56%
all others	4%

SYSE — Statistics & data analysis

SYSE postings	69%
all others	45%

BME — LabVIEW

BME postings	67%
all others	3%

Source: ECS 1100 Semiconductors Job Log · 180 postings from 15 employers, captured Sept 26, 2026

BME is based on 3 postings